SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I Veebala Jeeva (192526013 3) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V Jeeva

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	1	2	1	2	2
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	0	1	0	0

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SIMATS ENGINEERING

ASSESSMENT TOOL 2 — SCENARIO-BASED ASSIGNMENT

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

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Date: ___29/10/26_____ Signature: __v jeevan _____

This assignment contains worked solutions for all five scenarios in the supplied assessment document. Each scenario is developed across three pages, with calculations, explanations, verification, and final findings.

Code of Conduct

I certify that this submission is my work and that I have followed the academic integrity guidelines specified for this assessment. I understand that any violation may result in disciplinary action.

QUESTION 1 — SMART CAMPUS IPv4 SUBNET ALLOCATION (Page 1 of 3)

Scenario and objective. The Smart Campus has been allocated the IPv4 block **192.168.100.0/24**. The required departments are School of Computing (/25), School of Electronics (/26), and School of Mechanical Engineering (/27). The task is to determine each subnet's network address, broadcast address, valid host range, usable host count, and the unused range left in the original block.

1. Understanding the /24 address block

An IPv4 address contains 32 bits. The prefix /24 means that 24 bits identify the network and the remaining 8 bits identify host addresses. Therefore, the block 192.168.100.0/24 contains 2^8 = 256 total addresses, from 192.168.100.0 through 192.168.100.255. The first address is the network address and the last address is the broadcast address for the original /24 block.

A subnet mask is another way of describing the prefix. A /24 mask is 255.255.255.0. When a subnet uses a longer prefix, fewer bits remain for host addresses. A /25 has 7 host bits, a /26 has 6 host bits, and a /27 has 5 host bits. The number of usable host addresses in a normal IPv4 subnet is $2^h - 2$, where h is the number of host bits. The subtraction accounts for the network and broadcast addresses.

2. Allocation strategy

To avoid overlap, assign the largest subnet first. The /25 subnet needs 128 addresses, the /26 needs 64 addresses, and the /27 needs 32 addresses. Starting at the beginning of the /24 block, the aligned allocation is 192.168.100.0/25, then 192.168.100.128/26, then 192.168.100.192/27. Each next subnet begins on a valid block boundary and does not overlap the previous one.

A /25 has a block size of 256 – 128 = 128 in the last octet. A /26 has a block size of 64, and a /27 has a block size of 32. These block sizes help identify the valid network starts: .0 and .128 for /25; multiples of 64 for /26; and multiples of 32 for /27.

3. School of Computing — /25

Network address: **192.168.100.0**. Subnet mask: **255.255.255.128**. Since the block contains 128 addresses, the range is .0 to .127. The network address is .0 and the broadcast address is .127. Usable host addresses run from **192.168.100.1 to 192.168.100.126**. Host bits = 32 – 25 = 7; total addresses = 2^7 = 128; usable addresses = 128 – 2 = **126**.

QUESTION 1 — SMART CAMPUS IPv4 SUBNET ALLOCATION (Page 2 of 3)

4. School of Electronics — /26

After the first 128-address block, the next available aligned network begins at 192.168.100.128. Assign this range to the School of Electronics. Its network address is **192.168.100.128**, mask **255.255.255.192**, and block size 64. The subnet runs from .128 through .191, so the broadcast address is **192.168.100.191**. The usable host range is **192.168.100.129–192.168.100.190**.

The /26 prefix leaves 6 host bits: 32 – 26 = 6. Thus the subnet has 2^6 = 64 total addresses. Removing the network and broadcast addresses gives 64 – 2 = **62 usable host addresses**. The network address cannot be assigned to a computer because it identifies the subnet, and the broadcast address is reserved for sending traffic to all hosts in that subnet.

5. School of Mechanical Engineering — /27

The next unused aligned range begins at 192.168.100.192. Assign this range to the School of Mechanical Engineering using /27. Its subnet mask is **255.255.255.224**, and the block size is 32. The subnet range is .192 through .223. Therefore, the network address is **192.168.100.192**, broadcast address is **192.168.100.223**, and usable host range is **192.168.100.193–192.168.100.222**.

The number of host bits is 32 – 27 = 5. Total addresses = 2^5 = 32. Usable hosts = 32 – 2 = **30**. The chosen starting address .192 is a multiple of 32, so it is correctly aligned for a /27 subnet. The allocated subnet ends at .223, and the next address, .224, is outside this assigned subnet.

6. Combined allocation table

Department	Prefix / Mask	Network address	Usable host range	Broadcast	Usable hosts
Computing	/25\n255.255.255.128	192.168.100.0	192.168.100.1–.126	192.168.100.127	126
Electronics	/26\n255.255.255.192	192.168.100.128	192.168.100.129–.190	192.168.100.191	62
Mechanical	/27\n255.255.255.224	192.168.100.192	192.168.100.193–.222	192.168.100.223	30

The ranges are contiguous and non-overlapping: the /25 ends at .127, the /26 begins at .128 and ends at .191, and the /27 begins at .192 and ends at .223. This confirms that the allocation is valid.

QUESTION 1 — SMART CAMPUS IPv4 SUBNET ALLOCATION (Page 3 of 3)

7. Remaining unused IP address range

The original /24 block ends at 192.168.100.255. The last allocated subnet ends at 192.168.100.223. Therefore, the remaining unused range is **192.168.100.224–192.168.100.255**, which contains 32 addresses. This remaining range can be used for a future /27 subnet, because a /27 contains 32 addresses and the address .224 is aligned to a 32-address boundary.

The unused range .224–.255 includes the network address 192.168.100.224 and broadcast address 192.168.100.255 if it is later allocated as 192.168.100.224/27. Its usable hosts would then be 192.168.100.225–192.168.100.254, or 30 usable host addresses. Until that subnet is formally allocated, the entire range should be recorded as unassigned rather than assigned individually to devices.

8. Capacity check

Subnet	Total addresses	Reserved network/broadcast	Usable hosts
Computing /25	128	2	126
Electronics /26	64	2	62
Mechanical /27	32	2	30
Allocated total	224	6	218
Unused range	32	Not yet subnetted	Not yet assigned
Original /24 total	256		

The allocated subnet sizes add to 128 + 64 + 32 = 224 addresses. The original /24 has 256 addresses, so 256 − 224 = 32 addresses remain unused. This matches the range .224–.255 and confirms the arithmetic. Across the three assigned subnets, 126 + 62 + 30 = **218 usable host addresses** are available for endpoint devices.

9. Practical implementation recommendations

Configure the appropriate subnet mask on every endpoint in each department. Choose one usable host address in each subnet for the router interface or default gateway, and exclude that address from the automatic DHCP pool. For example, the campus could use 192.168.100.1 as the Computing gateway, 192.168.100.129 as the Electronics gateway, and 192.168.100.193 as the Mechanical gateway. These are examples only; the actual gateway plan should be documented and applied consistently.

Create separate DHCP scopes for the three subnets and ensure that each scope distributes the correct mask, default gateway, and DNS settings. Apply access-control rules between academic and administrative systems where required. After configuration, test that hosts can communicate within their own subnet, reach their gateway, and access permitted networks. Check that no two devices use the same IP address.

10. Final answer

Computing: network 192.168.100.0/25, broadcast 192.168.100.127, hosts 192.168.100.1–192.168.100.126, 126 usable hosts. Electronics: network 192.168.100.128/26, broadcast 192.168.100.191, hosts 192.168.100.129–192.168.100.190, 62 usable hosts. Mechanical: network 192.168.100.192/27, broadcast 192.168.100.223, hosts 192.168.100.193–192.168.100.222, 30 usable hosts. Unused range: **192.168.100.224–192.168.100.255** (32 addresses).

QUESTION 2 — MULTINATIONAL SOFTWARE COMPANY (Page 1 of 3)

Scenario. The company has the private IPv4 block **10.10.0.0/16**. Four departments have been assigned /24, /25, /26, and /27 subnets respectively. The aim is to determine the network and broadcast addresses, valid host ranges, usable host counts, and the unused range remaining in the /16 block.

1. Understanding the allocated block

A /16 prefix means that the first 16 bits identify the network and 16 bits remain for host addressing. The subnet mask is 255.255.0.0. The full address span is 10.10.0.0 through 10.10.255.255, giving 2^{16} = 65,536 total addresses. Because the company has many departments and may expand, only a small portion of this large block is needed for the four current assignments.

For a subnet with h host bits, the total number of addresses is 2^h. In conventional IPv4 subnetting, two addresses are reserved: the network address and directed broadcast address. Therefore usable hosts = 2^h – 2. The /24 has 8 host bits, /25 has 7, /26 has 6, and /27 has 5.

2. Software Development — /24

Assign 10.10.0.0/24 to Software Development. Its subnet mask is 255.255.255.0 and the block size is 256 addresses in the final octet. The network address is **10.10.0.0** and the broadcast address is **10.10.0.255**. Valid host addresses are **10.10.0.1–10.10.0.254**.

A /24 leaves 32 – 24 = 8 host bits. Total addresses = 2^8 = 256. After reserving the network and broadcast addresses, 256 – 2 = **254 usable host addresses** remain. This subnet is suitable for a department needing up to 254 ordinary IPv4 endpoint addresses, subject to reserving some addresses for gateways, servers, printers, or network devices.

3. Quality Assurance — /25

The next aligned subnet after 10.10.0.0/24 is 10.10.1.0/25. A /25 mask is 255.255.255.128 and the block size is 128 addresses. This subnet covers 10.10.1.0 through 10.10.1.127. The network address is **10.10.1.0**, broadcast is **10.10.1.127**, and usable hosts are **10.10.1.1–10.10.1.126**.

The /25 has 7 host bits, so it contains 2^7 = 128 total addresses and 128 – 2 = **126 usable hosts**. Starting it at .1.0 ensures that the subnet boundary is correctly aligned. It does not overlap the Software Development subnet, which ends at 10.10.0.255.

QUESTION 2 — MULTINATIONAL SOFTWARE COMPANY (Page 2 of 3)

4. Human Resources — /26

Allocate 10.10.1.128/26 to Human Resources. A /26 mask is 255.255.255.192, with a block size of 64 addresses. The range is 10.10.1.128 through 10.10.1.191. Network address: **10.10.1.128**. Broadcast address: **10.10.1.191**. Valid host range: **10.10.1.129–10.10.1.190**.

There are 6 host bits because 32 − 26 = 6. The subnet therefore contains 2^6 = 64 total addresses and 64 − 2 = **62 usable hosts**. The starting value 128 is a multiple of the /26 block size 64, so this is a valid network boundary.

5. Executive Management — /27

Allocate 10.10.1.192/27 to Executive Management. The subnet mask is 255.255.255.224 and the block size is 32 addresses. The subnet covers 10.10.1.192 through 10.10.1.223. Network address: **10.10.1.192**. Broadcast address: **10.10.1.223**. Valid hosts: **10.10.1.193–10.10.1.222**.

The /27 prefix leaves 5 host bits, giving 2^5 = 32 total addresses. Subtracting the network and broadcast addresses leaves **30 usable host addresses**. The network starts at .192, a multiple of 32, and the allocation follows directly after the Human Resources range without overlap.

6. Department allocation table

Department	Prefix / Mask	Network	Valid host range	Broadcast	Usable hosts
Software Development	/24\n255.255.255.0	10.10.0.0	10.10.0.1–10.10.0.254	10.10.0.255	254
Quality Assurance	/25\n255.255.255.128	10.10.1.0	10.10.1.1–10.10.1.126	10.10.1.127	126
Human Resources	/26\n255.255.255.192	10.10.1.128	10.10.1.129–10.10.1.190	10.10.1.191	62
Executive Management	/27\n255.255.255.224	10.10.1.192	10.10.1.193–10.10.1.222	10.10.1.223	30

These allocations are non-overlapping. Software Development occupies the 10.10.0.x range; the remaining three fit into successive parts of 10.10.1.x. Their network addresses respect the relevant block boundaries.

QUESTION 2 — MULTINATIONAL SOFTWARE COMPANY (Page 3 of 3)

7. Unused range remaining in the /16

The four assigned subnets occupy 256 + 128 + 64 + 32 = **480 total addresses**. The /16 block contains 65,536 addresses, so 65,536 – 480 = **65,056 addresses** remain unused. Under the allocation used here, the final assigned subnet ends at 10.10.1.223. The first unused address is therefore **10.10.1.224**, and the unused range continues through **10.10.255.255**.

The unused range is not one single conventional subnet with a single prefix as written above; it is simply the continuous remaining address interval after the assigned subnets. It can be divided later into aligned subnets of appropriate sizes. For example, 10.10.1.224–10.10.1.255 can form 10.10.1.224/27, while the rest of the /16 can be allocated to future departments using a documented subnet plan.

8. Address capacity verification

Subnet	Total addresses	Usable host addresses
Software Development /24	256	254
Quality Assurance /25	128	126
Human Resources /26	64	62
Executive Management /27	32	30
Total assigned	480	472
Total original /16	65,536	—
Unused addresses	65,056	Not yet allocated

The usable host totals add up to 254 + 126 + 62 + 30 = **472 usable host addresses**. The total assigned address count is 480 because each of the four subnets reserves its own network and broadcast address. This is why the total assigned addresses and total usable hosts differ by eight: two reserved addresses per subnet times four subnets.

9. Implementation and decision-making

Create a separate VLAN and IP subnet for each department where the network design supports it. Configure the router or Layer-3 switch with a gateway address from each subnet, and reserve those addresses so they are not assigned to endpoints by DHCP. Use separate DHCP scopes and verify that each scope uses the right subnet mask, gateway, and DNS server.

The unused /16 capacity should be preserved in an IP address management record. Allocate future blocks on correct subnet boundaries and avoid assigning a new subnet over an existing one. Use access-control lists or firewall rules to limit access to HR and executive systems when needed. Test local gateway reachability, inter-VLAN routing, DNS resolution, and approved access to internal services.

10. Final answer

Software Development: 10.10.0.0/24, broadcast 10.10.0.255, hosts 10.10.0.1–10.10.0.254, 254 usable. Quality Assurance: 10.10.1.0/25, broadcast 10.10.1.127, hosts 10.10.1.1–10.10.1.126, 126 usable. Human Resources: 10.10.1.128/26, broadcast 10.10.1.191, hosts 10.10.1.129–10.10.1.190, 62 usable. Executive Management: 10.10.1.192/27, broadcast 10.10.1.223, hosts 10.10.1.193–10.10.1.222, 30 usable. Remaining unused range: **10.10.1.224–10.10.255.255**, containing 65,056 addresses.

QUESTION 3 — SMART HOSPITAL IPv6 NETWORK (Page 1 of 3)

Scenario. A smart hospital is migrating from IPv4 to IPv6 for medical devices and IoT sensors. The assigned network is **2001:0db8:abcd:0000:0000:0000:0000:0000/64**. The required tasks are to compress the address, expand it back to full notation, identify the /64 network prefix, and calculate the number of addresses supported.

1. IPv6 address structure

An IPv6 address is 128 bits long and is written as eight groups called hextets. Each hextet contains four hexadecimal digits and represents 16 bits. Colons separate the eight hextets. The supplied address contains eight groups: 2001, 0db8, abcd, 0000, 0000, 0000, 0000, and 0000. A /64 prefix means the first 64 bits, or first four hextets, identify the network prefix; the remaining 64 bits are available for interface identifiers within the subnet.

IPv6 uses hexadecimal digits 0–9 and a–f. Unlike IPv4, IPv6 does not use a dotted-decimal subnet mask. Prefix length notation such as /64 indicates how many leading bits belong to the network. IPv6 addresses do not use a broadcast address; multicast and other IPv6 mechanisms are used for functions that would traditionally involve broadcast in IPv4.

2. Rules for compressing an IPv6 address

There are two main compression rules. First, leading zeroes within any individual hextet may be removed. For example, 0db8 can be written as db8, and 0000 can be written as 0. Trailing zeroes are not removed by this rule. Second, one consecutive sequence of all-zero hextets may be replaced by a double colon (::). The double colon can be used only once in an address, because using it more than once would make the number of omitted hextets ambiguous.

For the supplied address, the first hextet 2001 remains unchanged. The second hextet 0db8 becomes db8. The third hextet abcd remains unchanged. The last five hextets are all zero. Because these zero hextets are consecutive, they can be represented by one double colon.

3. Compressed notation

The full address **2001:0db8:abcd:0000:0000:0000:0000:0000** compresses to **2001:db8:abcd::**. The /64 prefix can be retained when showing the network, so the compressed network notation is **2001:db8:abcd::/64**. This is the standard shortened representation of the supplied network.

QUESTION 3 — SMART HOSPITAL IPv6 NETWORK (Page 2 of 3)

4. Expanding the compressed address

To expand 2001:db8:abcd::, restore each hextet to four hexadecimal digits and replace the double colon with the missing zero hextets. The visible hextets are 2001, db8, and abcd. The double colon represents five omitted hextets, because an IPv6 address must contain eight hextets in total and three are visible. Restore the five zero hextets as 0000.

The fully expanded address is **2001:0db8:abcd:0000:0000:0000:0000:0000**. This matches the original address exactly. When the /64 prefix is included, it is written as 2001:0db8:abcd:0000:0000:0000:0000:0000/64. Compression changes only the way the address is written; it does not change the underlying 128-bit value.

5. Identifying the /64 network prefix

A /64 prefix covers 64 bits. Each hextet contains 16 bits, so 64 bits equal four hextets. The network prefix is therefore the first four hextets: **2001:0db8:abcd:0000**. In compressed network form, this prefix is commonly represented as **2001:db8:abcd:0/64** or **2001:db8:abcd::/64**. The second form is the concise standard notation for the supplied network.

The remaining four hextets, or 64 bits, form the interface-identifier portion of the /64 subnet. A device interface in the network has the same first 64 prefix bits and a value in the remaining 64 bits. In practice, an IPv6 address may also be configured with privacy extensions or other interface-identifier methods, depending on the operating system and network policy.

6. Address range and total address count

There are 64 bits left after the /64 network prefix. Each remaining bit can be either 0 or 1, giving 2^{64} possible combinations. Therefore, the subnet contains **18,446,744,073,709,551,616 addresses** (approximately 1.8447×10^{19}). Unlike conventional IPv4 subnet calculations, there is no IPv6 broadcast address to subtract.

The address range starts with the all-zero interface identifier, 2001:db8:abcd::, and ends with all 64 interface bits set to one: 2001:db8:abcd:0:ffff:ffff:ffff:ffff. The all-zero address has special meaning in some IPv6 contexts, so address assignment and use should follow IPv6 rules and the hospital’s network configuration standards.

QUESTION 3 — SMART HOSPITAL IPv6 NETWORK (Page 3 of 3)

7. Calculation shown step by step

Step 1: IPv6 address length = 128 bits. Step 2: Network prefix length = 64 bits. Step 3: Host/interface bits remaining = 128 – 64 = 64 bits. Step 4: Total possible bit combinations = 2^{64}. Step 5: 2^{64} = 18,446,744,073,709,551,616. This is the total number of addresses in the /64 prefix.

Item	Answer
Original full address	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Compressed notation	2001:db8:abcd::/64
Expanded notation	2001:0db8:abcd:0000:0000:0000:0000:0000
Network prefix (64 bits)	2001:0db8:abcd:0000
Interface identifier length	64 bits
Total addresses	2^64 = 18,446,744,073,709,551,616

8. Practical implications for the hospital

A /64 is the standard subnet size for most IPv6 end-user networks and supports a very large address space for medical equipment, sensors, workstations, and infrastructure devices. The large number of possible addresses does not mean every address must be allocated manually. IPv6 supports automatic address configuration and other methods to simplify endpoint addressing. The hospital should choose a consistent addressing plan for wards, laboratories, imaging systems, administration, and IoT segments.

The network should be segmented according to operational and security requirements. For example, medical devices can be placed in separate VLANs or logical networks from visitor Wi-Fi and office computers. Routers and firewalls should permit only required flows, and monitoring should be enabled for device discovery and unusual traffic. IPv6 security policies should be configured explicitly rather than assuming that a large address space makes a network secure.

9. Important note about “host addresses”

The mathematical number of addresses in a /64 is 2^{64}. In real deployments, not every address is necessarily assigned to a device, and certain values may have special protocol significance. The assessment asks for the total possible host addresses; the direct prefix calculation is 2^{64}. Unlike IPv4, do not subtract two for a network and broadcast address. IPv6 does not define a broadcast address.

10. Final answer

Compressed address: **2001:db8:abcd::**; compressed network: **2001:db8:abcd::/64**. Expanded address: **2001:0db8:abcd:0000:0000:0000:0000:0000**. Network prefix: first 64 bits, **2001:0db8:abcd:0000**. Remaining interface bits: 64. Total addresses supported: **2^{64} = 18,446,744,073,709,551,616**.

QUESTION 4 — ISP ROUTER AND ROUTING TABLE (Page 1 of 3)

Scenario. An Internet Service Provider connects customer networks through a core router. The routing table contains 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24. A packet arrives with destination IP address **192.168.2.45**. The router must determine the destination subnet, matching route, next-hop interface or network, and the default route if no specific route matches.

1. What a routing table does

A router forwards IP packets between networks. It examines the destination IP address in each packet and compares that destination with entries in its routing table. A routing entry normally includes a destination prefix and information about how to reach it, such as an outgoing interface, a next-hop router, or both. The router selects the most specific matching route, which is known as longest-prefix matching.

A /24 route has a mask of 255.255.255.0. It matches all addresses whose first three octets are the same as the network address, while the final octet can range from 0 to 255. Therefore, 192.168.2.0/24 matches destination addresses from 192.168.2.0 through 192.168.2.255.

2. Examine the destination IP

The destination is **192.168.2.45**. Its first three octets are 192.168.2, which match the network 192.168.2.0/24. The final octet, 45, falls within the subnet's address range. The destination is not in 192.168.1.0/24 because that route requires the third octet to be 1. It is not in 192.168.3.0/24 because that route requires the third octet to be 3.

The packet's destination is a host address within 192.168.2.0/24. The network address is 192.168.2.0, the broadcast address is 192.168.2.255, and the valid host range is 192.168.2.1 through 192.168.2.254. The destination 192.168.2.45 is therefore a valid host address in that subnet.

3. Destination subnet

The destination subnet is **192.168.2.0/24**. This is the direct result of matching the packet's destination IP with the prefix listed in the routing table. The router should use the route for this subnet rather than routes for the 192.168.1.0/24 or 192.168.3.0/24 networks.

QUESTION 4 — ISP ROUTER AND ROUTING TABLE (Page 2 of 3)

4. Matching routing table entry

The matching route is **192.168.2.0/24**. Its subnet mask is 255.255.255.0, which means that the first 24 bits are compared. The destination 192.168.2.45 matches those first 24 bits exactly. Because all three given entries are /24 routes, there is no difference in prefix length among them; only the 192.168.2.0/24 entry matches this destination.

A simplified routing table can be shown as follows. The actual outgoing interface and next-hop address are not supplied in the question, so those values cannot be named precisely without additional topology information.

Destination prefix	Example matching range	Does 192.168.2.45 match?
192.168.1.0/24	192.168.1.0–192.168.1.255	No
192.168.2.0/24	192.168.2.0–192.168.2.255	Yes
192.168.3.0/24	192.168.3.0–192.168.3.255	No

5. Next-hop network/interface

The router forwards the packet using the forwarding information associated with 192.168.2.0/24. If the customer network is directly connected to the core router, the outgoing interface connected to that subnet is used, and the router resolves the destination's link-layer address as required. If the destination network is reachable through another router, the core router sends the packet to the configured next-hop router for that route.

The assessment does not provide interface names, router addresses, or a complete topology. Therefore, it is not possible to identify a specific interface such as GigabitEthernet0/1 or a specific next-hop IP address from the supplied information alone. The correct conceptual answer is: use the outgoing interface or next hop associated with the 192.168.2.0/24 routing-table entry.

6. Packet-forwarding sequence

First, the router receives the packet and reads destination 192.168.2.45. Second, it checks the routing table for matching prefixes. Third, it finds that 192.168.2.0/24 matches. Fourth, it selects the associated route and resolves the next-hop or directly connected destination. Fifth, it forwards the packet out the relevant interface. The router does not forward the packet to every network; it uses the route selected for the destination.

If the router is directly connected to the destination LAN, the router does not need another Layer-3 router as a next hop for that destination. It forwards on the local interface after resolving the destination's Layer-2 address. If the route is remote, the router uses the next-hop router address. This distinction depends on the route configuration and network topology.

QUESTION 4 — ISP ROUTER AND ROUTING TABLE (Page 3 of 3)

7. Default route

A default route is used when no more specific route in the routing table matches the destination and no other applicable route is available. In IPv4, the default route is written as **0.0.0.0/0**. It matches any IPv4 destination because the prefix length is zero. On a router, the default route normally points to an upstream next-hop router or an upstream interface.

For an ISP core router, the default route may point toward an upstream provider or the Internet backbone, depending on the network design. The question does not give the actual upstream router address or interface, so the exact next hop cannot be specified. If the destination does not match 192.168.1.0/24, 192.168.2.0/24, or 192.168.3.0/24, and a default route is configured, the router can use 0.0.0.0/0. If no matching route or default route exists, the router generally drops the packet and may send an ICMP destination-unreachable message, subject to configuration.

8. Summary table

Task	Answer
Destination IP	192.168.2.45
Destination subnet	192.168.2.0/24
Matching routing entry	192.168.2.0/24
Subnet mask	255.255.255.0
Valid host range of subnet	192.168.2.1–192.168.2.254
Next hop/interface	The interface or next hop configured for 192.168.2.0/24; exact value not supplied
Default route	0.0.0.0/0, if configured

9. Verification and operational considerations

A network engineer can verify the route with the router's route-table display command and check whether 192.168.2.0/24 is marked as directly connected or learned through a routing protocol. A ping or traceroute test can help verify reachability, but a failed ping alone does not prove that the route is missing because firewalls may block ICMP. The interface status and neighbor resolution should also be checked.

Routing policies should prevent loops and avoid unintended exposure between customer networks. The ISP should maintain accurate route advertisements and ensure that the route for each customer prefix points to the correct circuit or next-hop router. The default route should be installed only when the router has a valid upstream path. Route monitoring helps detect failures and incorrect announcements.

10. Final answer

The packet destination 192.168.2.45 belongs to **192.168.2.0/24**, so the matching entry is **192.168.2.0/24**. Forward it using the next hop or outgoing interface configured for that route. The exact interface/next-hop address is not provided in the scenario. The default IPv4 route is **0.0.0.0/0** and is used for destinations that have no more specific matching route, provided that the default route is configured.

QUESTION 5 — TWO OFFICE LANs THROUGH A LAYER-3 ROUTER (Page 1 of 3)

Scenario. A company connects Administration and Finance LANs through a Layer-3 router. Administration is 192.168.10.0/26 and uses gateway 192.168.10.1. Finance is 192.168.10.64/26 and uses gateway 192.168.10.65. The tasks are to calculate each LAN's network and broadcast addresses, valid host range, three example computer addresses, and default gateway.

1. Understanding the /26 subnet

The /26 prefix means 26 bits are used for the network and 6 bits remain for host addresses. The subnet mask is 255.255.255.192. The block size in the last octet is 256 − 192 = 64 addresses. Therefore /26 network boundaries in this /24 area occur at .0, .64, .128, and .192. The two supplied network addresses, .0 and .64, are correctly aligned.

Each /26 subnet contains 2^6 = 64 total addresses. In conventional IPv4 subnetting, the first address identifies the network and the final address is the broadcast address. The other 62 addresses can be used for host interfaces, subject to reserving addresses for gateways and infrastructure.

2. Administration LAN — 192.168.10.0/26

Network address: **192.168.10.0**. The block begins at .0 and includes 64 addresses, so it ends at .63. Broadcast address: **192.168.10.63**. The valid host range is **192.168.10.1–192.168.10.62**. Subnet mask: **255.255.255.192**. Total addresses = 64 and usable hosts = 64 − 2 = **62**.

The given Administration gateway, 192.168.10.1, is within the valid host range and is not the network or broadcast address. It is therefore a valid choice for the router interface serving Administration. Hosts on this LAN should use 192.168.10.1 as their default gateway to reach other IP networks, including the Finance LAN, if routing and security policy permit it.

3. Administration host examples

Three suitable computer addresses are **192.168.10.10**, **192.168.10.11**, and **192.168.10.12**, each with subnet mask 255.255.255.192 and default gateway 192.168.10.1. These addresses are within .1–.62 and do not conflict with the network address .0 or broadcast address .63. Before assignment, check the DHCP scope and address inventory to ensure that these addresses are not already in use.

QUESTION 5 — TWO OFFICE LANs THROUGH A LAYER-3 ROUTER (Page 2 of 3)

4. Finance LAN — 192.168.10.64/26

Finance begins at 192.168.10.64. With a block size of 64, the subnet spans .64 through .127. Network address: **192.168.10.64**. Broadcast address: **192.168.10.127**. Valid host range: **192.168.10.65–192.168.10.126**. The subnet mask is 255.255.255.192. Total addresses = 64 and usable hosts = **62**.

The supplied Finance gateway is 192.168.10.65, which is the first valid host address in this subnet. It is a valid host address and does not overlap with Administration's .0/26 subnet. Hosts on Finance should be configured to use 192.168.10.65 as their default gateway.

5. Finance host examples

Three suitable computer addresses are **192.168.10.70**, **192.168.10.71**, and **192.168.10.72**. Use subnet mask 255.255.255.192 and default gateway 192.168.10.65 for each. All three addresses fall between 192.168.10.65 and 192.168.10.126. They do not equal the network address .64 or broadcast address .127.

6. Combined LAN summary

Item	Administration LAN	Finance LAN
Network / prefix	192.168.10.0/26	192.168.10.64/26
Subnet mask	255.255.255.192	255.255.255.192
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1–192.168.10.62	192.168.10.65–192.168.10.126
Gateway	192.168.10.1	192.168.10.65
Usable host count	62	62

7. Why each LAN needs its own gateway

A host can communicate directly with devices on its own subnet using the local link. When it needs to reach a destination outside its subnet, it sends the packet to its configured default gateway. The gateway is the router interface that can route packets toward other networks. Administration's gateway must be an address in 192.168.10.0/26, while Finance's gateway must be in 192.168.10.64/26.

Because the two LANs use different /26 prefixes, hosts should not treat the other LAN as local. A host in Administration sends traffic for Finance to 192.168.10.1; the router then routes it toward the Finance network. A Finance host sends traffic for Administration to 192.168.10.65, and the router forwards it toward Administration. This assumes that the router has both interfaces or VLAN interfaces configured and that inter-LAN routing is enabled.

QUESTION 5 — TWO OFFICE LANs THROUGH A LAYER-3 ROUTER (Page 3 of 3)

8. Example host configuration table

LAN / device	IP address	Subnet mask	Default gateway
Administration PC 1	192.168.10.10	255.255.255.192	192.168.10.1
Administration PC 2	192.168.10.11	255.255.255.192	192.168.10.1
Administration PC 3	192.168.10.12	255.255.255.192	192.168.10.1
Finance PC 1	192.168.10.70	255.255.255.192	192.168.10.65
Finance PC 2	192.168.10.71	255.255.255.192	192.168.10.65
Finance PC 3	192.168.10.72	255.255.255.192	192.168.10.65

9. Routing and connectivity checks

First, verify that the router's Administration interface is configured as 192.168.10.1/26 and its Finance interface as 192.168.10.65/26. Confirm that both interfaces are operational. Second, verify that each PC has the correct address and /26 mask. Third, ping the local gateway from each PC. A successful ping confirms basic local connectivity to the router interface.

Next, test communication between a host in Administration and a host in Finance. The router must have both networks in its routing table; directly connected routes are normally added automatically when the interfaces are configured and up. If the ping fails, check interface status, VLAN membership, host firewall settings, gateway configuration, and any access-control rules between the two LANs.

Do not configure a computer with the subnet's network or broadcast address. Avoid duplicate addresses by documenting static allocations and configuring DHCP exclusions for gateways and manually assigned devices. For a production network, consider reserving addresses for printers, access points, servers, and management interfaces. Keep the IP address plan updated when devices are added or removed.

10. Final answer

Administration LAN: network **192.168.10.0/26**, broadcast **192.168.10.63**, valid hosts **192.168.10.1–192.168.10.62**, gateway **192.168.10.1**. Example PCs: 192.168.10.10, .11, and .12. Finance LAN: network **192.168.10.64/26**, broadcast **192.168.10.127**, valid hosts **192.168.10.65–192.168.10.126**, gateway **192.168.10.65**. Example PCs: 192.168.10.70, .71, and .72. Each subnet has 64 total addresses and 62 usable host addresses.

Conclusion

Correct subnetting separates the two departments into distinct broadcast domains, makes IP address management easier, and enables the Layer-3 router to control communication between them. Correct host addresses, subnet masks, and gateways are all necessary for reliable communication.